

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **embedded software modules** in **C/C++ and Python** for automotive ECU targets including **ARM microcontrollers** and **multi-core SoCs**, implementing feature requests and bug fixes with full **module testing** and **test report generation** aligned to **ASPICE** quality standards.
- Organized and executed **code reviews** of own and peer work products, ensuring compliance with **coding guidelines** and **QA requirements**; maintained all deliverables in **CM (Configuration Management) database** with proper **change control** and version management using **Git** and enterprise SCM tools.
- Supported embedded systems testing and **ECU configuration** across development teams, collaborated with cross-functional engineering groups on **test specifications** and **user specifications**, and participated in project and team meetings to report software development activities to team leadership.
- Managed **ticket generation and tracking** for defects and feature requests; implemented fixes and changes required by assigned tickets with turnaround **SLA compliance of 98%**, ensuring all modifications met **software quality processes** and company QA requirements.
- Applied understanding of **embedded development tools** including **emulators, debuggers, and profilers** to diagnose and resolve module-level issues; debugged microcontroller firmware and **multi-core SoC** interactions to achieve **30% reduction in field defect escape rate**.
- Contributed to development and maintenance of **software specifications** and technical documentation; ensured work products satisfied **ASPICE compliance** requirements and company software quality processes, supporting R&D; team in delivering automotive-grade embedded solutions.

SKILLS

Embedded Development Languages – C, C++, Python, ARM assembly.

Embedded Targets & Platforms – microcontrollers, DSP, ARM processors, multi-core SoCs, ECU development and configuration.

Development Tools & Debugging – emulators, debuggers, profilers, JTAG, oscilloscopes, embedded IDEs.

Version Control & Change Management – Git, Subversion, CM databases, configuration management, change control workflows, ASPICE compliance.

Software Quality & Testing – module testing, test report generation, code reviews, QA requirements, ASPICE standards, SLA management.

Automotive & Embedded Systems – ECU support and configuration, automotive software development, multi-core systems, real-time embedded constraints.

Collaboration & Reporting – cross-functional team meetings, ticket tracking systems, Jira, technical documentation, stakeholder reporting.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

PROJECTS

Embedded ECU Module Architecture & Multi-Core SoC Integration Framework (HCL Technologies)

- Designed and developed **embedded software modules in C/C++** for **ARM-based automotive ECUs** supporting **multi-core SoC** targets, with full **module test** harness and **test report generation** pipeline achieving **95% code coverage** against safety-critical requirements.
- Implemented **code review workflows** and **coding guidelines compliance** checks integrated into **Git** CI/CD pipeline; established CM database change-control processes reducing **configuration errors by 40%** and ensuring **ASPICE Level 3** process maturity.
- Deployed **embedded development tools** stack including **debuggers, emulators, and profilers** for cross-core synchronization issues; diagnosed and fixed microcontroller firmware deadlocks and SoC memory contention bugs, improving ECU stability from **92% to 98.5% uptime**.
- Conducted **ECU testing and configuration** support across 12-person R&D; team, coordinated **user specifications** and **test specifications** documentation with automotive domain engineers, and maintained **QA requirements** traceability reducing specification-defect cycle time by **35%**.
- Managed **ticket tracking** and defect resolution workflow; implemented **fixes and changes** for assigned tickets with **98% on-time closure rate**, prioritizing **ASPICE-compliant** quality processes and ensuring **SLA compliance** for critical embedded subsystems.

Automotive Software Quality & Embedded Test Automation Pipeline (HCL Technologies)

- Built integrated **module testing framework** in **C/C++** for microcontroller and **ARM processor** targets, automating **test report generation** and defect tracking to **Jira**; reduced manual test cycle time from **8 hours to 2 hours** per ECU configuration.
- Established **code review** governance and **coding guidelines** enforcement across embedded teams using **Git hooks and automated checkers**; achieved **zero critical defects in production** ECU releases over 18-month period through **ASPICE-aligned QA processes**.
- Supported **ECU configuration and testing** across multi-core embedded systems; debugged synchronization issues using **embedded development tools** and **profilers**, reducing microcontroller-level race conditions from **12 per release to 0**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)